

Amritaraj Nair

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EDUCATION

Texas A&M University — B.S. Computer Science Honors, Minor in Mathematics

Expected May 2029

Relevant Coursework: Data Structures & Algorithms, C++ Design, Discrete Structures, Engineering Calculus III, Python Programming

EXPERIENCE

Matic | Software Engineer Intern

Jun 2026 – Aug 2026

- Designed and shipped **Pulsematic**, an AI-powered inbox triage and auto-response drafting system for **athenaOne** and other **EHR inbox platforms**, saving physicians **over 90 minutes per provider/ 3 more patients seen per day**.
- Architected a multi-agent **orchestrator system on Google Cloud Platform (GCP) with Vertex AI** to route, prioritize, and optimize automated drafting across heterogeneous message types, improving response relevance and reducing manual edits.
- Built automated **CI/CD pipelines with Google Cloud Build triggers**, streamlining deployment velocity and reducing release turnaround time across services.
- Engineered a **PHI-compliant data management pipeline secured via Firestore**, enforcing strict access controls and data handling safeguards for protected health information at scale.

MagNet Agents | Software Engineer Intern

May 2025 – Aug 2025

- Engineered **data pipelines and web scraping systems** to extract and structure high-value legal lead data from public sources, enabling automated client acquisition workflows.
- Developed **AI-powered outreach generation features** using LLMs to create personalized messaging and legal opportunity summaries, improving engagement and conversion.
- Built core **backend systems for lead matching and tracking**, connecting lawyer profiles with relevant prospects and supporting a platform generating **\$50K+ ARR**.
- Collaborated in a fast-paced startup environment to ship **end-to-end product features**, contributing to a Cornell-backed platform at the intersection of legal tech and AI.

AI Modeling for Bio-Economic Systems | Undergraduate Research

Aug 2025 – Present

- Applied **agent-based modeling (Mesa)** and machine learning to optimize livestock nutrition under economic and environmental constraints.
- Migrated legacy **R workflows to Python**, improving reproducibility, scalability, and cross-pipeline integration.
- Built cross-platform simulation systems in **Python, Java, and NetLogo** to evaluate sustainability and production outcomes.

Maroon Fund — Scholars of Finance | Quantitative Developer

Feb 2025 – Present

- Developed **quantitative models and screening frameworks** for equity research across a **\$70K+ student-managed fund**.
- Engineered **data pipelines and valuation tools** using financial datasets to improve investment analysis efficiency.
- Automated **research workflows**, delivering data-driven insights for stock pitch processes and portfolio decisions.

PROJECTS

AlphaForge | Founder & Solo Developer

March 2026

- Built **AlphaForge**, a full-stack quantitative investing platform enabling users to design, test, and deploy trading strategies.
- Developed a multi-level **Quant IDE** with a Scratch-like builder for beginners and an advanced coding environment supporting multi-language strategies.
- Implemented **backtesting and paper trading systems**, improving strategy validation and iteration efficiency.
- Engineered a real-time **market tracker** covering stocks, ETFs, indices, and crypto with integrated analytics and financial news.
- Designed an interactive **learning platform** to explain financial metrics directly within the user interface.
- Integrated **Yahoo Finance, Alpaca, and Seeking Alpha APIs**; deployed on **Vercel** for scalable, low-latency performance.

Shot Sensei | Lead Developer & Co-Founder

Mar 2026 – Present

- Built a computer vision hackathon project for competitive and training-based pickleball, winning the **Hookem Hacks 2026 (Startup Ready Award)**.
- Developed a real-time gameplay mode where users play against an AI bot, alongside a training mode using **Gemini** and **ElevenLabs** to analyze and coach all pickleball strokes (serve, volley, forehand, backhand).
- Implemented pose and shot detection using **OpenCV** and **YOLOv8** for stroke classification and performance feedback.
- Engineered backend data infrastructure with **Supabase** to track player win/loss records and shot-level scoring analytics.
- Invited back to the **McCombs School of Business at UT Austin** to pitch Shot Sensei to **Pear VC (Khalil Fuller)**, presenting it as both a competitive AI-powered pickleball game playable anytime, anywhere and a personalized training system.

ClinicalHours | CTO & Lead Engineer

Dec 2025 – Present

- Built an **AI-powered email and call automation system** using **Gmail API** and **GoHighLevel (GHL)** to act as a virtual receptionist, scheduling meetings with clinics and accelerating application workflows for pre-med students.
- Integrated **MapBox, Google APIs, and Resend API** to enable geolocation-based clinic discovery and automated communication pipelines.
- Processed large-scale **U.S. hospital datasets** in Python, transforming raw data into structured, production-ready formats.
- Implemented premium features like AI resume and application tools using **OpenAI and Gemini**. Partnered with clinics like **BCS Free Health Clinic**.

TECHNICAL SKILLS & AWARDS

Languages: Python, C++, Java, TypeScript, SQL

Frameworks: React, Flask, FastAPI, Pandas, NumPy, Matplotlib

Tools: Git, Docker, AWS EC2/IAM, Google Cloud, Supabase, CI/CD

Awards: UT Austin Hook'em Hacks 2026 **Winner** (Most Startup Ready + Multimodal Track) · TidalTAMU Google Gemini Track **1st Place** (2026) · Outstanding Undergraduate Researcher Award · President's Endowed Scholar · Google Labs Makeathon · FRC 4192 Impact Award